Air Defense System

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MINI LAB PROJECT REPORT

This Report Presented in Partial Fulfillment of the course

CSE412: Computer Graphics & Lab in the Computer Science and Engineering Department

DAFFODIL INTERNATIONAL UNIVERSITY

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DECLARATION

We hereby declare that this lab project has been done by us under the supervision of **Md. Mahabul Alom Santo**, **Lecturer**, Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere as lab projects.

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COURSE & PROGRAM OUTCOME

The following course have course outcomes as following:

Table 1: Course Outcome Statements

CO's	Statements
CO1	Identify, define, and relate fundamental computer graphics components such as primitives, shapes, objects, transformation functions, and raster algorithms required to construct a 2D animated scene
CO2	Apply knowledge of OpenGL, GLUT, and essential graphics programming techniques (drawing primitives, color models, coordinate systems) to implement visually accurate objects in the Air defence system environment.
CO3	Analyze and design the scene structure using modular functions, graphical modeling concepts, and algorithmic breakdowns (like Bresenham line-drawing) to represent different components of the simulation.
CO4	Develop and implement a complete animated graphics system using OpenGL, integrating real-time animation, user interaction, and classic algorithms (Bresenham) to create an effective and academically justified solution.

Table 2: Mapping of CO, PO, Blooms, KP and CEP

CO	PO	Blooms	KP	CEP
CO1	PO1	C1, C2	KP3	EP1, EP3
CO2	PO2	C2	KP3	EP1, EP3
CO3	PO3	C4, A1	KP3	EP1, EP2
CO4	PO3	C3, C6, A3, P3	KP4	EP1, EP3

The mapping justification of this table is provided in section **4.3.1**, **4.3.2,** and **4.3.3**.

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Chapter 1

Introduction

1.1 Introduction

Computer Graphics is widely used to create visual scenes, animations, and interactive simulations using mathematical formulas and graphical primitives. This project, titled "Air Defence System – Animated Missile Interception using OpenGL", focuses on building a dynamic 2D defense environment where multiple objects—tank, fighter plane, missiles, cars, and city background—are drawn and animated in real-time.

A special highlight of this project is the implementation of a state-based battle system, where the simulation operates through different stages such as Awaiting Launch, Intercept in Progress, Plane Destroyed, *and* Tank Destroyed. Additionally, the system introduces an interactive launch feature using a clickable button, which allows the user to control the missile firing process.

This makes the project not only visually engaging but also conceptually informative.

1.2 Motivation

The main motivation for this project was to explore how real-time animation, graphical primitives, and state-based logic can be combined to create an interactive and visually meaningful simulation in Computer Graphics. Instead of developing only static 2D scenes, this project aims to demonstrate dynamic movement, missile interception, and user interaction within a single graphical environment.To overcome the limitations of static 2D scenes

- To overcome the limitations of static 2D graphical scenes
- To understand state-based animation and real-time system behavior
- To improve visualization of missile trajectories and interception process
- To provide an interactive and centralized simulation environment
- To enhance the overall learning experience in computer graphics

1.3 Objectives

The primary objective of this project is to design and implement a complete animated 2D air defence simulation using OpenGL, where real-time animation, state-based logic, and graphical transformations are demonstrated in an interactive way. This project aims to combine multiple Computer Graphics concepts—object modeling, animation, missile interception, and user interaction—within a single system.The specific objectives of the project are:

- To develop a visually realistic and animated 2D air defence environment
- To implement Bresenham's Line Algorithm
- To enable an interactive launch feature using mouse input
- To apply graphical transformations such as translation
- To use polygons, circles, and color shading
- To demonstrate real-time rendering and double buffering
- To create an educational, easy-to-understand demonstration

1.4 Feasibility Study

The feasibility study for this project evaluates whether the proposed animated air defence simulation—built using OpenGL, C++ programming, and real-time animation techniques—can be developed and executed smoothly within the available technical, operational, and economic constraints. After analysis, the project is found to be highly feasible due to its manageable complexity, low resource requirements, and strong educational value.

1.5 Gap Analysis

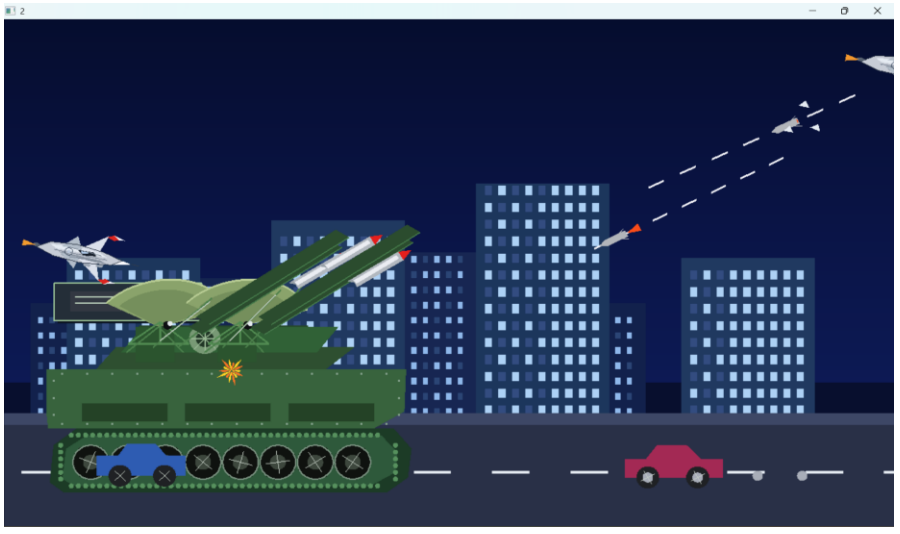
This gap analysis identifies shortcomings commonly found in beginner and intermediate computer-graphics projects and explains how this Air Defence System project addresses each gap. Each gap is described clearly, along with the specific design or code approach that solves it, and any remaining limitations with suggestions for future improvement.

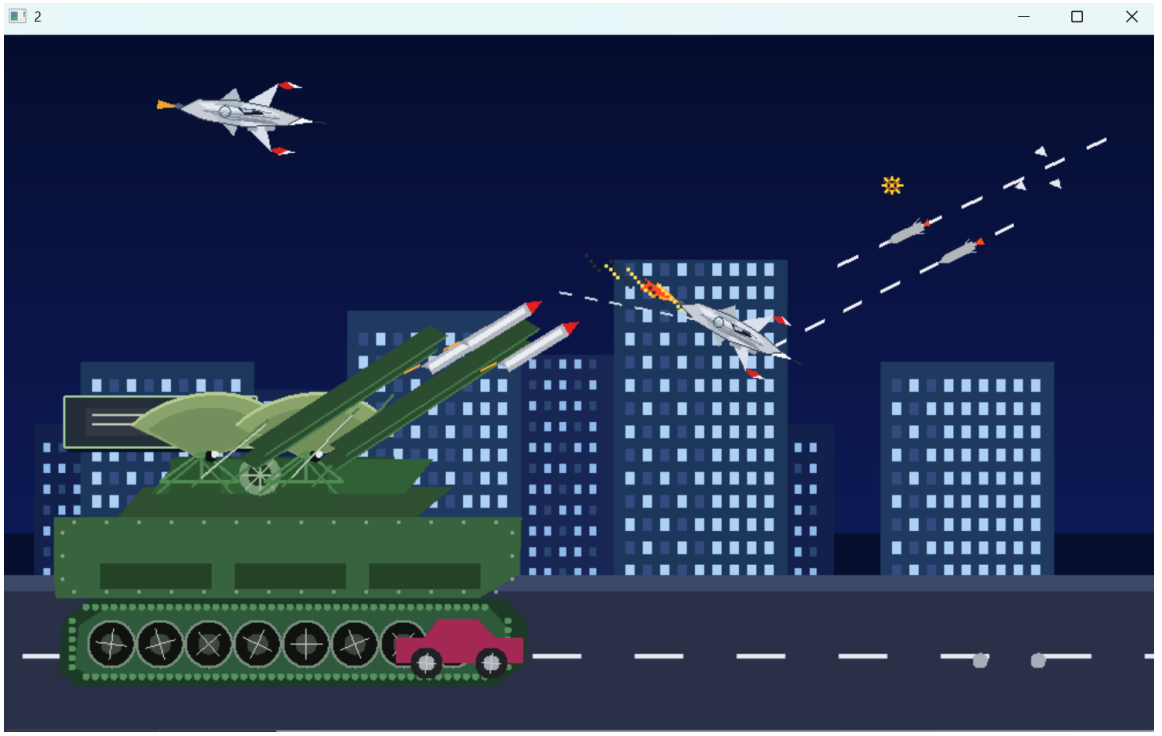
- **No use of real-time system logic:** Most projects only display simple animations without any structured control system.
  - **Our Solution:** This project uses a complete state-based system (Awaiting Launch, Intercept in Progress, Plane Destroyed, Tank Destroyed) to control the entire simulation dynamically.
- **No interaction or comparison:** Many projects do not allow user involvement in the simulation.
  - **Our Solution:** An interactive launch button is implemented, allowing users to trigger missile interception using mouse input, making the system responsive and engaging.
- **Very simple scenes:** Some projects contain only one or two moving objects.
  - **Our Solution:** This project includes multiple animated elements tank, fighter plane, missiles, cars, and city environment to create a complete and dynamic scene..
  - **Lack of modular and clear structure:** Many projects place all logic in a single function.
  - **Our Solution:** The project is divided into multiple functions such as drawing tank, plane, missiles, and handling animation logic, making the code clean and maintainable.

1.6 Project Outcome

The Air Defence System project successfully created a complete and animated 2D simulation using OpenGL. The project combined multiple Computer Graphics concepts such as object modeling, real-time animation, user interaction, and state-based system design to produce a visually rich and educational output.

The main outcome of the project is a smooth, realistic, and interactive defence environment where a tank, fighter plane, missiles, vehicles, and city background are drawn and animated continuously. The implementation of a state-based battle system adds strong conceptual value, allowing users to control missile launching through an interactive button and observe different outcomes such as interception, plane destruction, or tank destruction.

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Chapter 2

Proposed Methodology/Architecture

This chapter explains how the **RoadPulse** system is designed, how the components work together, and how the animation, shapes, and Bresenham algorithm are implemented. The methodology follows a step-by-step structured approach from requirement analysis to final scene rendering.

2.1 Requirement Analysis & Design Specification

The project requirements are divided into functional and non-functional parts.

Functional Requirements:

- Draw all objects in the scene (defense base, missile launcher/tank, enemy aircraft/missile, sky, radar, explosion effects, ground elements).
- Animate moving objects such as missiles, enemy targets (aircraft), and radar scanning movement.
- Implement missile launching and target tracking mechanism using transformation and motion logic.
- Provide user control system (keyboard input) to launch missiles or control movement.
- Update the scene continuously using the display loop for real-time animation.
- Support keyboard interactions for controlling defense system actions (e.g., fire, move, toggle features).

Non-Functional Requirements:

- Render smoothly without flickering using double buffering.
- Maintain clean and modular code structure for easy understanding and modification.
- Ensure low computational cost using efficient drawing techniques.
- Display all graphical objects clearly with proper color and positioning.
- Run efficiently on standard computers without requiring high-end hardware.

Design Specification

The design of this project focuses on integrating graphical rendering, animation control, and interactive simulation using OpenGL. The system is structured modularly so that each component works independently while contributing to a unified animation.

Architecture: Layered Graphics Rendering Structure

The project follows a layered design:

- Rendering Layer – draws all objects (defense base, missiles, aircraft, radar, sky, explosion effects).
- Animation Layer – controls movement of missiles, enemy objects, and radar rotation.
- Logic Layer – handles missile targeting, collision detection, and movement calculation.
- Interaction Layer – manages keyboard inputs (missile launch, movement control, feature toggles).

- **Graphics Design (Object-Based Scene Construction)**

All visual elements are created using basic geometric primitives:

- Polygons for base structures, vehicles, and aircraft bodies
- Circles for radar, explosion effects, and rotating elements
- Lines for missile paths, targeting indicators, and radar scan
- Color shading to represent different objects (sky, ground, fire, explosion)
- Transformations (glTranslatef, glRotatef) to animate movements
- Bresenham's algorithms for road

- **Animation Design: Real-Time Transformation System**

  The animation system updates object positions every frame:

  - ✓ Missile → moves toward target using incremental position updates
  - ✓ Enemy aircraft → moves across the screen continuously
  - ✓ Radar → rotates using angular transformation
  - ✓ Explosion → appears dynamically on collision
  - ✓ Scene refreshes using `glutPostRedisplay()`

- **Algorithm Design: Bresenham's Line Drawing Module**

  The center lane marker uses:

  - Bresenham's Line Algorithm for pixel-perfect dashed lines
  - Translation-based motion for missile and enemy movement
  - Basic collision detection between missile and target
  - Conditional logic for triggering explosion animation
  - Real-time input handling for user actions

- **UI/UX: Visual Scene Layout**

  Although this is not a web UI, the visual layout is carefully designed:

  - Clear sky and ground separation
  - Defense system placed strategically on ground
  - Enemy objects appear from realistic directions
  - Missile trajectory is visible for understanding movement
  - Smooth animation for better user experience

2.1.1 Overview

The project is a 2D animated air defense simulation developed using OpenGL. It integrates computer graphics concepts such as object rendering, real-time animation, and user interaction within a single system.

The system demonstrates a dynamic environment where a defense unit launches missiles to intercept moving enemy targets. Various components such as radar, missiles, aircraft, and explosion effects are combined to create a complete simulation.

A key feature of the project is the interactive control system, allowing users to trigger actions such as missile launching and observe real-time movement and collision behavior.

This project demonstrates how rendering, animation, and logical interaction work together to simulate a real-world defense scenario. It is designed for educational purposes, helping learners understand animation techniques, transformation, and basic simulation logic in computer graphics.

2.1.2 Proposed Methodology/ System Design

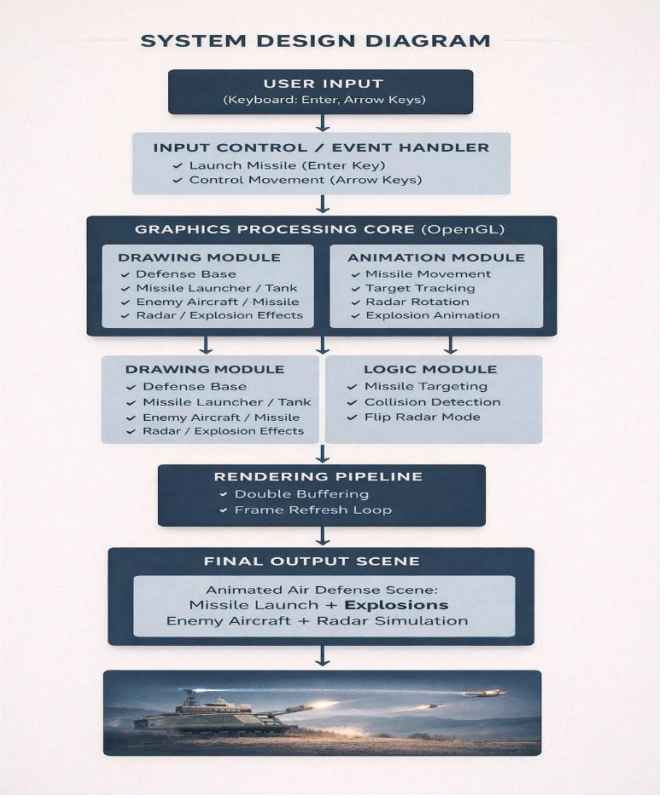
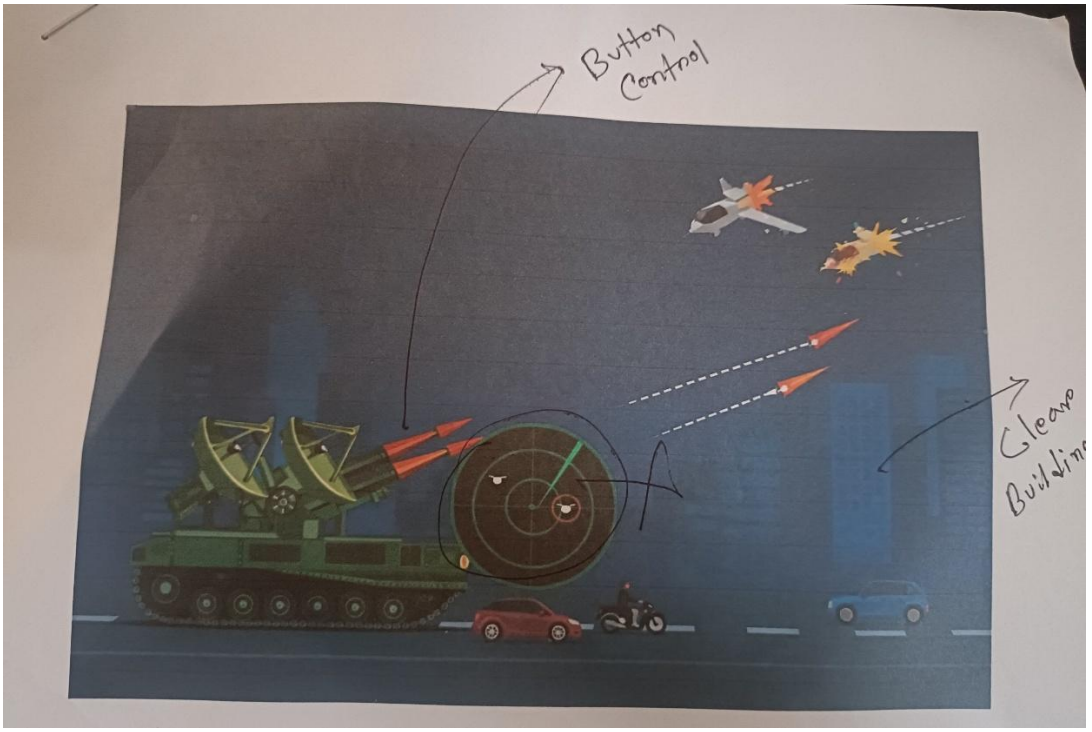


Figure 0-1: System Design Diagram of the RoadPulse Graphics Project

Proposed Methodology/ System Design



2.2 Overall Project Plan

The overall project plan for the Air Defense System followed a structured approach from initial concept to final implementation. The development process began with analyzing the project requirements and identifying how computer graphics concepts such as object rendering, animation, and interaction could be integrated into a real-time simulation. The focus was on designing a dynamic system where a defense unit can detect and intercept enemy targets using animated missile behavior.

During the design phase, the system was divided into modular components such as defense base, missile launcher, enemy aircraft, radar system, and explosion effects. Each component was planned separately to ensure clarity and maintainability of the code. The animation logic and object interactions were also designed at this stage, including missile movement, target tracking, and collision handling.

In the implementation phase, OpenGL primitives were used to construct all graphical objects. Transformation functions were applied to animate missiles, enemy aircraft, and radar rotation. Interactive features were added using keyboard controls to allow users to launch missiles and control system behavior. The logic for collision detection and explosion effects was also implemented to simulate realistic interception scenarios.

After completing development, the system was tested thoroughly to ensure smooth animation, correct object movement, accurate collision detection, and proper response to user inputs. Performance was evaluated to confirm that the program runs efficiently without lag or rendering issues.

Finally, documentation, code refinement, and demonstration preparation were carried out to make the project easy to understand, present, and evaluate. This structured workflow ensured that the Air Defense System successfully met both functional and academic objectives while maintaining clarity, interactivity, and visual effectiveness.

Chapter 3

Implementation and Results

The Air Defense System project was implemented using C, OpenGL, and GLUT, focusing on 2D computer graphics rendering, real-time animation, and interactive simulation techniques. The final output presents an animated defense environment featuring a missile launcher, moving enemy aircraft, radar system, and explosion effects. The system demonstrates dynamic missile launching, target tracking, and collision-based interactions within a visually structured scene. The implementation successfully achieved smooth animation, modular rendering of different components, and clear visualization of movement and interaction logic, making the project both technically effective and educational.

Implementation

The implementation of RoadPulse followed a structured and incremental development process. Work The implementation of the Air Defense System followed a structured and incremental development process. The work began by analyzing the overall system requirements and identifying the core components needed to simulate a real-time defense environment. The project focused on building a dynamic scene that includes a defense base, missile launcher, enemy aircraft, radar system, and explosion effects, ensuring that each element contributes to a realistic and interactive simulation.

The development process started with setting up the OpenGL environment using GLUT and configuring a 2D orthographic projection to ensure consistent rendering across different devices. The scene was divided into modular drawing functions such as defenseBase(), missileLauncher(), enemyAircraft(), radar(), and explosionEffect(). Each module was implemented using basic OpenGL primitives like GL_POLYGON, GL_POINTS, and GL_LINES, allowing clear and efficient construction of graphical objects.

A key part of the implementation was the development of missile movement and target interaction logic. Missile trajectories were controlled using translation transformations, enabling missiles to move smoothly toward enemy targets. Basic collision detection logic was implemented to identify when a missile intersects with an enemy object, triggering explosion effects. This added a realistic interaction between moving objects within the system.

Animation was achieved through continuous updates of object positions using transformation functions such as glTranslatef() and glRotatef(). Enemy aircraft move across the screen, radar components rotate to simulate scanning, and missiles travel dynamically toward targets. The scene is refreshed continuously using GLUT's display callback function, ensuring real-time animation. Double buffering was used to eliminate flickering and provide smooth visual output.

Interactive features were implemented using keyboard handlers (glutKeyboardFunc and glutSpecialFunc). These controls allow users to launch missiles, adjust movements, and interact with the simulation environment in real time. This interaction enhances the usability and demonstration capability of the project.

Extensive testing was conducted to ensure that all modules function correctly, animations remain smooth, and user inputs are properly handled. The system was verified for accurate missile movement, correct collision detection, and proper rendering of explosion effects. Overall, the implementation successfully transformed the conceptual design into a fully functional animated air defense simulation system with interactive and real-time graphical behavior

3.1 Performance Analysis

Performance analysis focused on evaluating rendering efficiency, animation smoothness, input responsiveness, and interaction accuracy within the Air Defense System. The project demonstrated stable real-time animation, with missiles, enemy aircraft, and radar components moving smoothly without noticeable frame drops. Rendering operations such as drawing the defense base, aircraft, radar, and explosion effects were efficiently handled using lightweight OpenGL primitives.

The movement and interaction logic performed accurately throughout the simulation. Missile trajectories followed smooth paths toward targets, and collision detection correctly triggered explosion effects at the appropriate positions. The system responded instantly to user inputs such as missile launch and control commands, confirming effective keyboard event handling.

Load testing was conducted by increasing animation speed and the number of active objects in the scene. The program remained responsive and maintained consistent frame rates without performance degradation. Continuous animation loops were executed without memory leaks or graphical glitches, ensuring system stability over extended runtime.

Additionally, input controls responded immediately when the OpenGL window was active, allowing seamless interaction with the simulation. The synchronization between animation, rendering, and user input ensured a smooth and realistic experience.

Overall, the system demonstrated strong performance with efficient rendering, smooth animation, accurate interaction logic, and reliable responsiveness—successfully meeting the expected technical standards of a 2D computer graphics simulation.

3.2 Results and Discussion

The results of the Air Defense System project confirmed that the system successfully met all major objectives defined during the design and requirement phases. The animated environment effectively integrates multiple graphical components such as the defense base, missile launcher, enemy aircraft, radar system, and explosion effects into a visually coherent and interactive simulation.

The implementation of missile movement and collision detection logic played a key role in achieving the project goals. Missile trajectories were displayed clearly, and collision events between missiles and enemy targets triggered accurate explosion effects, demonstrating the interaction between moving objects in a graphical environment. This feature proved highly effective for both demonstration and evaluation purposes, as it clearly illustrates real-time animation and interaction logic.

Feedback from testing highlighted the smoothness of the animation, the responsiveness of user controls, and the clarity of the visual elements. The system remained stable during continuous execution, including repeated missile launches and ongoing enemy movement. The modular design of functions also contributed to better code organization, making the system easier to understand, maintain, and extend.

Although the project achieved its primary objectives, several improvements can be considered for future development. These include adding advanced visual effects such as dynamic lighting or sound integration, implementing more sophisticated collision detection techniques, and introducing additional features like multiple enemy types or automated defense logic.

Overall, the Air Defense System stands as a complete and effective demonstration of 2D computer graphics, real-time animation, and interactive simulation, fulfilling both technical and academic requirements.

Chapter 4

Engineering Standards and Mapping

The Air Defense System adheres to computer graphics development standards, structured programming practices, and efficient rendering techniques. The project aligns with engineering principles such as modular system design, real-time animation control, interaction handling, and performance optimization. The use of OpenGL ensures proper graphical rendering standards, while the implementation of movement logic and collision detection demonstrates practical application of core computer graphics and simulation concepts.

4.1 Impact on Society, Environment and Sustainability

Although primarily an academic graphics project, the Air Defense System demonstrates how simulation-based visualization can contribute to education, awareness, and technological understanding. By simulating a defense scenario, the project highlights how graphical systems can represent real-world situations in a controlled and interactive digital environment. Since the system is fully software-based, it does not consume physical resources, making it environmentally friendly.Impact on Life

RoadPulse helps students and learners understand core principles of computer graphics, animation, and raster algorithms. By visually demonstrating how scenes are constructed, how objects move, and how lines are drawn pixel-by-pixel, it enhances conceptual clarity and practical learning. Such systems encourage experimentation, creativity, and problem-solving—skills essential for academic and professional growth in engineering fields.

4.1.1 Impact on Society & Environment

From a societal perspective, the Air Defense System serves as an educational simulation tool that can be used in academic environments to demonstrate graphical modeling and animation techniques. It also shows how simulation can be applied in areas such as defense training, monitoring systems, and visualization technologies.

Environmentally, the project promotes digital learning and simulation-based experimentation, reducing the need for physical models or resources and supporting sustainable technological practices.

Ethical Aspects

The project follows ethical computing practices by ensuring that:

- All development is done using open-source tools (OpenGL, GLUT), supporting accessible learning
- No harmful, misleading, or inappropriate content is included
- The system does not collect or store any user data, ensuring privacy

4.1.2 Sustainability Plan

T he Air Defense System is designed using a modular structure, making it easy to update, extend, and reuse in future projects. Features such as animation control, interaction handling, and collision logic can be enhanced further to develop more complex simulations.

Sustainability is ensured by:

- ✓ Clean and maintainable code structure
- ✓ Compatibility with standard systems and platforms
- ✓ Reusability for future academic and development purposes
- ✓ Potential expansion with advanced features such as AI-based targeting, multiple defense units, or enhanced graphical effects

4.2 Project Management and Team Work

Although implemented individually, the project followed a structured development approach similar to professional engineering workflows. The process included requirement analysis, system design, implementation, testing, and documentation.

Tasks such as module creation, animation development, interaction handling, and debugging were completed in a step-by-step manner to ensure clarity and progress. Proper planning ensured that the project met academic requirements within the given timeline. Documentation of each component made the system easier to understand, evaluate, and improve.

4.3 Complex Engineering Problem

Developing the Air Defense System presented a complex engineering problem involving the integration of multiple graphical components, animation behaviors, and real-time interaction within a single system. The challenge was to simulate a dynamic defense environment where missiles, enemy aircraft, radar systems, and explosion effects operate together smoothly.

A major complexity was maintaining synchronization between multiple moving objects within a fixed 2D coordinate system while ensuring smooth animation and stable performance. Each component missile launcher, enemy targets, radar, and visual effects was implemented as a separate module but needed to function cohesively in a unified scene.

Another significant challenge was implementing real-time interaction, where user inputs trigger missile launches and system responses without interrupting the animation flow. This required careful handling of input events, animation updates, and rendering cycles to avoid flickering or delays.

Additionally, designing collision detection and explosion effects required precise control of object positions and timing to ensure realistic simulation behavior.

Overall, the project successfully addressed these engineering challenges by combining modular design, efficient rendering, and interactive control into a stable and functional graphical system.

4.3.1 Mapping of Program Outcome

In this section, the complex graphics problem addressed in Air defence system—integrating classical raster algorithms, animation techniques, modular scene rendering, and user interaction—is mapped to the targeted **Program Outcomes (POs)**. The justification explains how each outcome is achieved through the design and implementation of the project.

Table 4-1: Justification of Program Outcomes

PO's	Justification
Engineering Knowledge	Applied foundational knowledge of computer graphics, OpenGL primitives, coordinate systems, mathematical modeling, and raster algorithms to design and implement a fully functional animated environment. Demonstrated the ability to integrate classical algorithms (Bresenham line-drawing) with real-time rendering and animation techniques.
Problem Analysis	Identified and formulated solutions for a complex graphical problem involving the synchronization of multiple animated objects, modular scene construction, and algorithmic accuracy. Analyzed limitations in the original scene and implemented a Bresenham-based improvement to enhance realism while ensuring optimal performance in the rendering pipeline.
Design/Development of Solutions	Designed and developed a complete 2D animated air defense simulation that balances visual quality, system performance, and user interaction. Implemented modular drawing functions for components such as the defense base, missile launcher, enemy aircraft, and radar system, along with interaction-driven logic for missile movement and collision detection. Integrated real-time animation and keyboard-controlled actions to ensure accurate simulation behavior,usability, and responsive system performance.

4.3.2 Complex Problem Solving

The Air defence systems project involves solving a complex engineering problem that integrates classical raster algorithms, modular scene construction, and real-time animation within a unified graphics system. The problem-solving process aligns with several **Knowledge Profiles (KP)** defined in engineering education. Each mapping below explains how these knowledge profiles were applied and why they were essential for the project.

Table 4-2: Mapping with complex problem solving.

EP1 Dept of Knowledge	**EP2** Range of Conflicting Requirement s	**EP3** Depth of Analysis	**EP4** Familiarity of Issues	**EP5** Extent of Applicable Codes	**EP6** Extent of Stakeholder Involvement	**EP7** Inter-dependence
✓	✓	✗	✓	✓	✗	✗

4.3.3 Engineering Activities

In this section, provide a mapping with engineering activities. For each mapping add subsections to put rationale (Use Table 4-3).

Table 4-3: Mapping with complex engineering activities.

EA1 Range of resources	**EA2** Level of Interaction	**EA3** Innovation	**EA4** Consequences for society and environment	**EA5** Familiarity
✓		✗	✓	✓

Chapter 5

Conclusion

RoadPulse successfully demonstrates how classical computer graphics techniques, real-time animation, and algorithmic drawing can be combined to create an interactive and visually appealing traffic simulation. The project highlights how fundamental graphics knowledge and efficient rendering practices can be applied to develop an educational and engaging animated environment.

5.1 Summary

This project focused on designing and developing an Air Defense System — a 2D animated simulation built using OpenGL. The main objective was to create a dynamic graphical environment featuring a defense base, missile launcher, enemy aircraft, radar system, and explosion effects, while incorporating real-time interaction and movement logic.

The system was structured into modular components for drawing the defense base, missiles, enemy targets, radar, and environment elements. A key feature of the project is the implementation of missile movement and collision detection, which enables realistic interaction between defense and enemy objects. Keyboard-based controls allow users to launch missiles and interact with the simulation, demonstrating real-time responsiveness and system control.

5.2 Limitation

Although Air defence systems performs effectively, several limitations remain:

- **2D Only:** The system is limited to 2D rendering and does not include advanced 3D transformations or realistic lighting models.
- **No Texture or Shader Support:** All objects are created using basic OpenGL primitives, without advanced textures or shading techniques**.**
- **Simple Collision Logic:** Collision detection is basic and may not handle complex or highly dynamic scenarios.
- **Limited Interaction:** User interaction is restricted to basic keyboard controls; more advanced controls could improve usability.
- **Fixed Behavior Patterns:** Enemy movement and missile behavior follow predefined paths without intelligent or adaptive logic**.**

5.3 Future Work

Several improvements can be made to expand RoadPulse and make it more realistic, interactive, and visually appealing.

The following future work items would significantly enhance the project:

- **Add Advanced Visual Effects:** Implement dynamic lighting, explosion animations, and improved graphical effects.
- **Introduce Textures & Shaders:** Apply texture mapping and shading techniques to create more realistic objects.
- **Implement 3D Environment:** Extend the project into a 3D simulation using perspective projection and depth.
- **AI-Based Targeting System:** Add intelligent behavior for enemy movement and automated defense responses.
- **Enhanced User Interaction:** Include more controls such as aiming, speed adjustment, and multiple missile systems.
- **Sound Integration:** Add sound effects such as missile launch, explosion, and radar signals.
- **Advanced Simulation Features:** Introduce multiple enemy types, scoring systems, and real-time difficulty adjustment.

Source codes link: **https://github.com/himel509/Missile-Defense-Simulation-using-OpenGL-C-.git**

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